Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.03	20.17	
worker nodes in service, mean	6.00	6.00	-0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.00	2.02	+0.7%
power (W), mean	628	630	+0.3%
power (W), peak	651	664	+2.1%
energy (Wh)	210	212	+1.0%
energy, parked workers still on at idle power (Wh)	210	212	+1.0%
CPU used (cores), mean	0.853	0.902	+5.7%
CPU allocatable (cores), mean	24.00	24.00	-0.0%
utilisation (used / allocatable)	0.036	0.038	+5.7%
energy per core-hour (Wh)	736	698	-5.1%
node-hours per core-hour	7.03	6.65	-5.4%
pending pods, pod-minutes	0.250	0.800	+220.0%
pending pods, peak	1.00	3.00	+200.0%
HPA replicas, mean	9.06	9.00	-0.7%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.500	0.783	+56.7%
requests timed	1,070	1,150	+7.5%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	126	46.15	-63.5%
response time (ms), median	113	40.70	-63.9%
response time (ms), 95th percentile	263	69.07	-73.8%
response time (ms), 99th percentile	376	82.90	-77.9%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.7464	7.6437	-1.3%	[-2.2797, +2.0645]	STEADY, no significant difference
kwh_per_core_hour	0.8066	0.7964	-1.3%	[-0.2154, +0.2007]	STEADY, no significant difference
utilisation	0.0351	0.0372	+5.9%	[-0.0084, +0.0132]	STEADY, no significant difference
pending_pod_minutes_per_hour	0.7143	2.3040	+222.6%	[-2.1429, +6.9120]	STEADY, no significant difference
hpa_shortfall_minutes_per_hour	1.4343	2.2560	+57.3%	[-2.1006, +3.7440]	STEADY, no significant difference

What Omni-Compass did

• decisions: 120 (one per minute)

• HPA target writes: 0

• node-pool resizes: 3

• scheduling-floor adds (pending pods): 0

• power-cap pod resizes (in place): 0

• rollout actions: 4

• power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	617	614	5.25	4.00	0.000	0.000	0.553	0.487
1	6.00	6.00	616	615	6.00	4.50	0.000	0.000	0.553	0.521
2	6.00	6.00	616	616	6.00	6.00	0.000	0.000	0.550	0.560
3	6.00	6.00	620	622	6.00	6.75	0.000	0.000	0.652	0.673
4	6.00	6.00	632	634	8.50	9.25	0.250	0.750	0.967	1.00
5	6.00	6.00	632	633	10.00	10.00	0.000	0.000	0.953	0.996
6	6.00	6.00	633	642	10.00	10.00	0.000	0.000	1.01	1.22
7	6.00	6.00	645	653	10.00	10.00	0.000	0.000	1.29	1.53
8	6.00	6.00	647	647	10.00	10.00	0.000	0.000	1.36	1.37
9	6.00	6.00	646	658	10.00	10.00	0.000	0.000	1.35	1.66
10	6.00	6.00	641	653	10.00	10.00	0.000	0.000	1.22	1.53
11	6.00	6.00	616	615	10.00	10.00	0.000	0.000	0.553	0.517
12	6.00	6.00	614	618	10.00	10.00	0.000	0.000	0.489	0.589
13	6.00	6.00	620	624	10.00	10.00	0.000	0.000	0.626	0.758
14	6.00	6.00	632	632	10.00	10.00	0.000	0.000	0.951	0.975
15	6.00	6.00	632	634	10.00	10.00	0.000	0.000	0.950	1.02
16	6.00	6.00	632	631	10.00	10.00	0.000	0.000	0.969	0.957
17	6.00	6.00	624	627	10.00	10.00	0.000	0.000	0.755	0.848
18	6.00	6.00	618	613	10.00	10.00	0.000	0.000	0.577	0.455
19	6.00	6.00	618	616	10.00	10.00	0.000	0.000	0.603	0.544

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.